

Data Preparation & Feature Scaling Lab (Practical)

Welcome to the practical lab for Data Preparation & Feature Scaling! In this session, you will learn how to load tabular data using **Pandas**, clean missing values (both numerical and categorical) using **Scikit-Learn's SimpleImputer**, convert text categories into numbers using **One-Hot Encoding**, and standardize feature scales using **StandardScaler** to prepare your data for machine learning models.

1. Concept Overview

In this lab, you will:

- Load a dataset from a CSV file into a Pandas DataFrame and inspect its structure.
- Handle missing or null values gracefully using imputation techniques (Mean for numbers, Mode for categories).
- Transform categorical text data into machine-readable numerical formats using One-Hot Encoding.
- Scale numerical features so that variables with larger ranges (e.g., square footage in thousands) do not overpower smaller ranges (e.g., number of rooms).

2. Step-by-Step Lab Instructions

Step 1: Create a Sample Dataset (`dataset.csv`)

Create a new directory named `ml_project` and a file named `dataset.csv` inside it. Add some sample data with intentional missing values (`NaN` or empty cells) and a categorical column (`house_type`) to simulate a real-world dataset:

```
csv
sqrft,rooms,house_type,target_price
1500,3,Villa,250000
2000,4,Apartment,320000
,3,Villa,210000
1200,,Condo,180000
```

2500,5,,400000
1800,3,Apartment,290000

Step 2: Create a Data Preparation Script (`prep_data.py`)

Create a Python script named `prep_data.py` in your project folder to load, clean, encode, and scale the data:

```
# prep_data.py
import pandas as pd
import numpy as np
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler, OneHotEncoder

# 1. Load the dataset using Pandas
print("--- 1. Loading Dataset ---")
df = pd.read_csv("dataset.csv")
print(df)
print("\nMissing values before cleaning:\n", df.isnull().sum())

# Separate features (X) and target (y)
X = df[["sqft", "rooms", "house_type"]]
y = df["target_price"]

# Separate numerical and categorical columns for processing
num_cols = ["sqft", "rooms"]
cat_cols = ["house_type"]

# 2. Handle missing values (Imputation)
print("\n--- 2. Handling Missing Values ---")
# Numerical Imputer: replace missing with column mean
num_imputer = SimpleImputer(strategy="mean")
X_num_imputed = num_imputer.fit_transform(X[num_cols])

# Categorical Imputer: replace missing with the most frequent value (mode)
cat_imputer = SimpleImputer(strategy="most_frequent")
X_cat_imputed = cat_imputer.fit_transform(X[cat_cols])
print("Imputed Categorical Feature:\n", X_cat_imputed)

# 3. Encode Categorical Data (One-Hot Encoding)
print("\n--- 3. Encoding Categorical Data ---")
encoder = OneHotEncoder(sparse_output=False) # sparse_output=False returns a standard array
X_cat_encoded = encoder.fit_transform(X_cat_imputed)
print("One-Hot Encoded Categories:\n", X_cat_encoded)
print("Category Names:", encoder.get_feature_names_out(cat_cols))

# 4. Scale Numerical Features (StandardScaler)
print("\n--- 4. Scaling Numerical Features ---")
# StandardScaler shifts data to mean=0 and variance=1.
```

```
# (Alternatively, MinMaxScaler scales data between 0 and 1)
scaler = StandardScaler()
X_num_scaled = scaler.fit_transform(X_num_imputed)
print("Scaled Numerical Array:\n", X_num_scaled)

# 5. Combine Processed Features
print("\n--- 5. Final Processed Dataset ---")
# Concatenate the scaled numerical features and encoded categorical
features horizontally
X_final = np.hstack((X_num_scaled, X_cat_encoded))
print(X_final)
```

Step 3: Run Your Script

Open your terminal, ensure your virtual environment is active, and run the script:

```
python prep_data.py
```

Observe how Pandas identifies the missing values across both data types. The script automatically handles the empty cells, converts the text categories (`Villa` , `Apartment` , `Condo`) into binary flags, and scales the large square footage numbers down to standardized distributions.

3. Reference Links & Further Reading

- **Pandas IO Tools Documentation:** Guide on reading CSV, Excel, and SQL data into DataFrames.
 - **Scikit-Learn Preprocessing Guide:** Official documentation on imputation, scaling, and normalization.
 - **Scikit-Learn Categorical Encoding:** A deep dive into `OneHotEncoder` and `OrdinalEncoder` .
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